

Article

A Comparison between Powder Bed Fusion of Polyamide 12 and Aluminum Computer Numeric Control Machining: A Carbon Footprint and Energy Assessment

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Abstract: This study presents a comparative life cycle assessment (LCA) of two manufacturing scenarios for a camera housing: Powder Bed Fusion (PBF) using polyamide 12 (PA-12) and Computer Numeric Control (CNC) machining using aluminum, for a cradle-to-gate boundary. The selected impact categories were cumulative energy demand (CED) and global warming potential (GWP). The key findings indicate that the PA-12 PBF part outperformed the aluminum CNC machining one in terms of environmental and energy performance, showing a significant reduction of approximately 90% in equivalent carbon emissions and 84% in cumulative energy demand. Sensitivity analyses revealed that the PBF process was highly sensitive to changes in the proportion of virgin/recycled material for the printing process, variations in the life cycle inventory (LCI) data sources for PA-12 powder production, and changes in the transportation system for imported materials, as is the case for the main input in the process (PA-12 powder). Sensitivity analysis also showed less impact for the PBF camera housing even considering a lifespan of one-fifth that of the aluminum for the impact categories considered. However, it should be noted that this analysis did not include considerations for the usage and end-of-life phases, which may have significant contributions to the overall environmental impact.

Keywords: additive manufacturing; multi-jet fusion; metal to plastic; LCA; carbon footprint; energy demand



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1. Introduction

The use of polymers as a replacement for metal has emerged as a significant trend in various industries. With advancements in polymer materials and manufacturing techniques, designers are exploring the potential of replacing traditional metal components with polymers for structural and non-structural purposes. Examples include the use of fiber-reinforced polymers in aerospace and automotive applications [1], high-performance polymers such as polyetheretherketone (PEEK) in oil and gas and automotive structural components [2,3], and thermoplastics to replace non-structural components in marine engines [4]. This shift is primarily driven by weight and cost reduction, but also by improved design flexibility and functionality, which has become even more relevant following the rise of additive manufacturing (AM) [5].

AM, commonly known as 3D printing, is the process of adding materials, typically layer by layer, in order to manufacture an object from a 3D computer model [6,7]. Compared to conventional manufacturing processes, it allows for greater design freedom, rapid production of small quantities at relatively low costs, and higher material efficiency use [8,9].

Although there are still certain reservations about its applicability for mass production, it is considered a key component of Industry 4.0 [10]. In addition to economic and design factors, studies also suggest that AM has the potential to bring environmental benefits to the production chain of products when compared to conventional manufacturing. This includes reducing raw materials and energy consumption, transportation distances between the manufacturer and consumers, and waste generation [11,12] in comparison with conventional processes.

In this context, the Life Cycle Assessment (LCA) can be used to quantify and compare the environmental performance of products. The LCA methodology can estimate environmental impacts by compiling the inputs and outputs of material and energy resources related to a product throughout all its life cycle stages [13,14], and, since manufacturing impacts can vary considerably depending on factors such as the raw material and specific machine used in the process, product geometry, machine use profile, and production volume, the LCA proves to be an essential step in decision-making towards sustainability [15].

For instance, Faludi et al. [16] compared the sustainability of Acrylonitrile Butadiene Styrene (ABS) parts using a Computer Numeric Control (CNC) machining mill and two different methods of additive manufacturing: Fused Deposition Modelling (FDM) and Polyjet. Environmental impacts were quantified considering the ReCiPe Endpoint H method, assuming a cradle-to-grave boundary. The results showed that FDM had the lowest environmental impacts compared to the other options; however, they were highly sensitive to the machine used and assumptions related to the use phase (rate of utilization during the year). Garcia et al. [17] compared plastic parts (ABS) using FDM and injection molding to determine the cradle-to-gate environmental impacts in terms of global warming potential (GWP) and cumulative energy demand (CED). The results showed that FDM had lower impacts for the production of up to 14 parts, but the injection molding process generated less impact for batch sizes above 50 parts. London et al. [18] assessed the cradle-to-gate life cycle CED and GWP of the multi-jet fusion technology and injection molding process. The functional unit considered was 1 kg of a PA-12 product, considering production quantities of up to 100,000 plastic enclosure parts. The authors observed that energy consumption and emissions remained lower than injection molding for a production quantity of up to 700 parts. LCA has been extensively used to compare additive and conventional manufacturing, especially of polymers [18–21] and metals [22–24]. However, there is still a gap in information regarding the comparison of impacts for cases of metal replacement with a polymer component.

This study considered a case where an aluminum camera housing produced by CNC machining was replaced by a redesigned polyamide 12 (PA-12) part produced using PBF, an additive manufacturing process. The objective was to compare the lifecycle environmental impacts of both products through a cradle-to-gate LCA study to identify the main hotspots and to provide Life Cycle Inventory (LCI) data for the PBF process, which are relatively scarce in the literature.

2. Materials and Methods

2.1. Camera Housing

This comparative LCA study focuses on the manufacturing of a camera housing prototype, part of a robotic tool designed for subsea operations. This particular component was designed to hold a MultiSense S7 camera in a watertight chamber with a support structure for a BlueView M900 sonar device and to withstand pressures up to 6 bars. It was originally manufactured in lightweight aluminum alloy (AA6061) by CNC machining (Figure 1a), following the material used for the structure of the robotic tool. Seeking to reduce its overall weight, a metal-to-plastic replacement was proposed using a PBF process that could print parts in PA-12, polyamide 11 (PA-11), and polypropylene (PP). Considering the requirement of the part being watertight, PA-12 was chosen as the best option due to its superior finish, allowing for smaller clearances at the joints. The original design was

also modified to reduce the number of components and connections while fulfilling the mechanical requirements considering the material change (Figure 1b). The main properties of the parts produced in both processes can be found in Table 1.

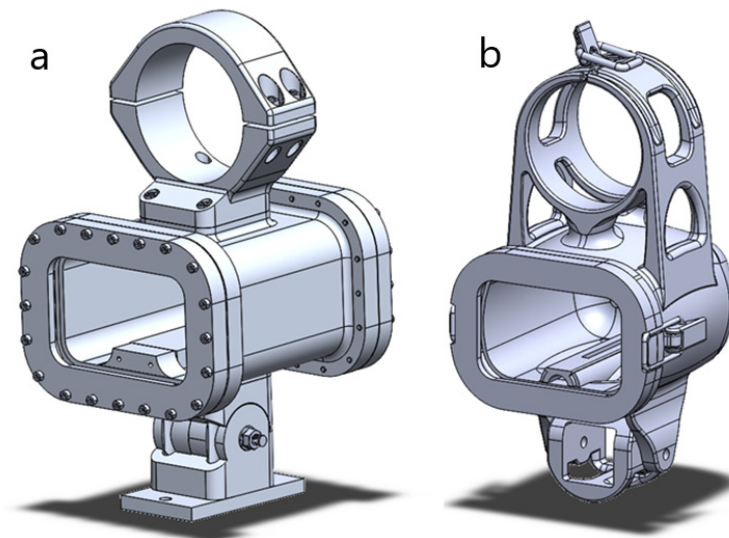


Figure 1. 3D models of the camera housing produced using (a) CNC machining and (b) PBF.

Table 1. Properties of camera housing manufactured in different processes.

Properties	Unit	Manufacturing Process	
		CNC Machining	PBF
Materials	-	Aluminum 6061	PA-12 powder
Weight	kg	6.40	1.84
Estimated lifespan	year	5	1
Production time per part	hour	~54	~12

2.2. Scope Definition

A comparative LCA was conducted based on ISO 14040 [13] and ISO 14044 [25] guidelines. The declared unit of the product system was one camera housing, with the characteristics described in the previous section, and with a lifespan of 5 years. Initially, given the absence of data for PA-12 durability in subsea conditions, a reference flow of one part for both manufacturing options was considered. However, a sensitivity analysis was conducted assuming different lifespan alternatives for the PA-12 part. The modeling approach was attributional, and the study boundary extended from cradle to gate.

The foreground and background boundaries used to produce the camera housing in PA-12 and aluminum can be seen in Figure 2. The foreground boundary refers to the flows and processes that can be directly controlled and for which there was primary data available, while the background represents the flows and processes related to the supply chain of inputs, use phase, and treatment and disposal of wastes.

2.3. Life Cycle Inventory (LCI)

The foreground life cycle inventory was based on primary data from reports and technical documentation, while the background processes used information from the Ecoinvent database version 3.8 [26]. A cut-off rule of 1% was applied to the total inputs and outputs of mass and energy.

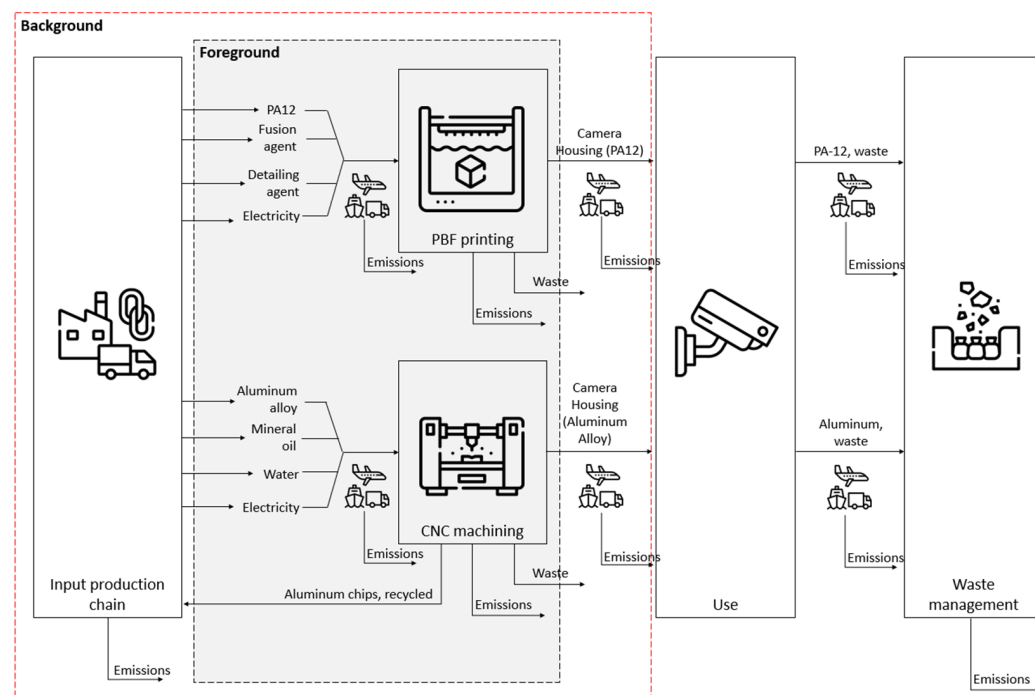


Figure 2. Product system of the camera housing production from cradle to gate.

2.3.1. Powder Bed Fusion (PBF) Additive Manufacturing Process

The PA-12 camera housing was printed by an HP Multi-Jet Fusion (HP MJF) 5210 printer at the Additive Manufacturing Bureau of SENAI CIMATEC. The printer has a maximum chamber capacity of 41 L (380 mm × 284 mm × 380 mm) and uses PA-12 powder, fusion, detailing agents, and electricity as main inputs. The camera housing required a build height of 287.06 mm (around 75.5% of total volume capacity) with a packing density (volume of printed material by total chamber building volume) of 5.21% when printed alone. However, since the packing density can range between 8% and 12% without compromising printing quality [27,28], we modeled the PBF process for a 10% packing density and assumed a criterion for the allocation of impact burdens based on the mass of the printed parts.

The quantity of PA-12 powder used in the process was estimated based on the printing chamber volume of ~31 L (380 mm × 284 mm × 287.06 mm) and a material density of 1.01 g/cm³. Life cycle impact assessment (LCIA) data for PA-12 powder production were only found in one report from Fraunhofer EMI [29]; however, since the results were not peer-reviewed and did not provide information about data quality, this study opted for using LCIA data for nylon 6 instead, assuming negligible differences in production [30,31]. The MJF printer can use up to 80% of recycled PA-12 powder from previous printing jobs in the process; therefore, impact burdens for the PA-12 powder production only consider the input of virgin material (20%).

The consumption of fusion and detailing agents per kg of printed material were estimated based on data collected from 10 printing jobs provided by the Bureau. Background inventory regarding the production of these agents followed the same method adopted by [18], and can be seen in Tables A1 and A2.

Electricity use was estimated based on the power consumption of the printer multiplied by the time necessary for each process step (Table A3). Data on the time required for powder preparation and loading into the equipment, preparation and cooling time during printing, and unpacking of the printed part, were extracted from the HP MJF 5210 User Guide [32] and the operational assumptions described by London [33]. Data on equipment power consumption were obtained from technical documents (manuals, user guides) (Appendix A).

The transportation inventory for inputs can be seen in Table A4. We assumed that the PA-12 powder was produced in the city of Marl, DE, while the fusion and detailing agents were produced in Corvalis, EUA [18].

The plastic waste generated during the process was assumed to be managed through landfilling. Additionally, the printed parts undergo a cleaning process through sandblasting, which requires electricity and generates plastic waste (around 10% of the total part weight). A comprehensive overview of the inputs and outputs associated with the PBF process can be found in Tables 2 and 3.

Table 2. Inventory of the production of one camera housing, without post-processing, in PA-12 PBF.

Item	Value	Unit	SD ² _g	Source
Input				
Nylon 6	6.25	kg	1.306	(1)
HP fusion agent	0.10	kg	1.306	(1); (2)
HP detailing agent	0.07	kg	1.306	(1); (2)
Electricity	110.74	kWh	1.306	(3); (4)
Transport, truck	27.54	t.km	2.029	(2); (5)
Transport, container ship	81.93	t.km	2.029	(2); (6)
Output				
Reference product				
Camera housing, PA-12, without post-processing	2.02	kg	-	(1)
Co-products				
Printed PA-12 parts	1.76	kg		(1)
Waste for treatment				
Inert solid waste	2.47	kg	1.306	(1)

(1) SENAI CIMATEC [34]; (2) London et al. [18]; (3) HP [32]; (4) London et al. [33]; (5) Google Maps [35]; (6) Seagate [36].

Table 3. Inventory of the production of one camera housing, post-processed, in PA-12 PBF, cradle-to-gate lifecycle.

Item	Value	Unit	SD ² _g	Source
Input				
Camera housing, PA-12, without post-processing	2.02	kg	1.306	-
Electricity	1.15	kWh	1.306	(1); (2)
Output				
Reference product				
Camera housing, PA-12, processed	1	Un.	-	
Waste for treatment				
Inert solid waste	0.18	kg	1.306	

(1) SENAI CIMATEC [34]; (2) London et al. [18].

2.3.2. CNC Machining

The aluminum camera housing was manufactured using a Romi Discovery 4022 CNC machining center at SENAI CIMATEC, located in Salvador, BA. The data collection occurred post-production and relied primarily on technical reports and internal information. The inputs to the process included aluminum blocks (type AA 6061), semi-synthetic lubricant oil, and electricity. Throughout the machining process, waste is generated in the form of aluminum chips and waste oil. These aluminum chips, along with other metal waste generated from various machining jobs, are combined and collected by a recycling company for proper disposal and recycling. The “at point of substitution” (APOS) approach was used to establish the system boundary concerning the recycling of the aluminum chips.

The total quantity of aluminum was determined by directly measuring the final part and evaluating the number of aluminum blocks required, considering an approximate machining allowance of 15 mm. A total of five blocks were utilized during the machining process, with an estimated combined weight of 28 kg, assuming a density of 2.71 g/cm³.

The lubricating oil consumption was estimated based on Faludi et al. [16] and the Ecoinvent 3.8 datasets for the aluminum machining of large parts. LCIA data on background processes were acquired from the Ecoinvent database.

As there was no available direct data for electricity consumption, an estimation was made by referencing the findings of Montes [37]. In this study, the energy consumption during the use phase of a Romy model D800 station was monitored for each task performed. By analyzing the results, an average power consumption was estimated for the equipment setup and for the actual machining step. The duration of each step was estimated through internal communication with the technical team responsible for the project at the time (refer to Table A5 for details).

The transportation inventory for the machining process can be seen in Table A6. We assumed that all the inputs were produced in São Paulo, BR, and transported by truck to Salvador, BR.

The life cycle inventory to produce one camera housing in aluminum can be seen in Table 4.

Table 4. Inventory of the production of one camera housing, in aluminum alloy CNC machining, cradle-to-gate lifecycle.

Item	Value	Unit	SD ² _g	Source
Input				
Aluminum, wrought alloy	28.00	kg	1.102	(1)
Lubricating oil	0.08	kg	1.102	(2)
Electricity	183.20	kWh	1.102	(1); (3)
Transport, truck	55.52	t.km	1.419	(4)
Output				
Reference product				
Camera housing, aluminum	1	Un.	-	
Waste for treatment				
Aluminum scrap	21.60	kg	1.102	Estimated
Waste mineral oil	0.08	kg	1.099	Mass balance

(1) SENAI CIMATEC [34]; (2) ecoinvent 3.8 dataset “aluminium milling, large parts | aluminium removed by milling, large parts | APOS, U—RoW”; (3) Montes [37]; (4) Google Maps [35].

2.3.3. Uncertainty Assessment

The uncertainty of the data used for the foreground inventory flows was characterized using the uncertainty factors [38] from the Pedigree Matrix method [39]. The method consists of a qualitative assessment of data sources, which includes a basic uncertainty based on the type of input and output, as well as additional uncertainty factors for five data characteristics: reliability, completeness, temporal correlation, geographic correlation, and further technological correlation. The uncertainty factors are then combined using Equation 1 to calculate a squared geometric standard deviation (SD_g²).

$$SD_g^2 = \exp^{\sqrt{[\ln(U_b)]^2 + [\ln(U_1)]^2 + [\ln(U_2)]^2 + [\ln(U_3)]^2 + [\ln(U_4)]^2 + [\ln(U_5)]^2}} \quad (1)$$

where U_b is the basic uncertainty factor, and U_i represents the uncertainty factors for reliability (U_1), completeness (U_2), temporal correlation (U_3), geographic correlation (U_4), and technological correlation (U_5), respectively.

2.4. Life Cycle Impact Assessment (LCIA)

The product system was modeled using the software OpenLCA version 2.0 with the Ecoinvent database version 3.8 using the APOS (At Point Of Substitution) library [35]. To account for uncertainty, the Montecarlo simulation was applied with 10000 iterations and a 95% confidence interval. This methodology facilitated the propagation of uncertainties from both background and foreground inventory data to the life cycle impact assessment

(LCIA) results, resulting in a more comprehensive and reliable analysis of the system's environmental impact.

The assessed categories in this study were cumulative energy demand (CED), which quantifies the primary energy demand connected within the production, use, and disposal of a product [40], and global warming potential (GWP), which measures the amount of energy that the greenhouse emissions from the lifecycle of a product will absorb in the atmosphere over a certain time period [41]. The characterization methods used were Cumulative Energy Demand version 1.11 [42] for CED, and IPCC 2021 100a [43] for GWP.

These categories were selected because they are the most affected by the environmental mechanisms of the two product systems. In the case of the PA-12 housing, the production of plastic is a major source of greenhouse gas emissions and non-renewable fuel use. In 2015, plastic production accounted for 4.5% of global emissions or 1.9 gigatons of CO₂ equivalent. Almost half of these emissions (44%) were related to coal-based energy and heat supply for production [44]. Additionally, the high energy consumption of the printing process contributes significantly to the environmental impact of additive manufacturing [16,17,45,46].

On the other hand, for the aluminum housing, the aluminum industry is highly energy intensive, has a significant environmental impact in terms of carbon emissions, and releases a large proportion of energy as waste heat [47]. Additionally, being a subtractive manufacturing process, machining has a significant burden contribution related to waste material management [16].

2.5. Sensitivity Analysis

Sensitivity analyses (SA) were conducted to investigate the effect of different assumptions on the PA-12 housing manufacturing:

- SA1: Change in the percentage of recycled PA-12 used to fill the printing chamber.

In the PBF process, the manufacturer recommends using a powder mixture of 20% virgin and 80% recycled PA-12 from previous processes. This assumption was used as the basis for modeling the inputs in this study. In order to explore opportunities for higher rates of recycled powder use, a sensitivity analysis was conducted considering a proportion of 90% recycled material.

- SA2: Use of available LCIA data on the production of PA-12 powder.

Data on the life cycle impact assessment (LCIA) of PA-12 powder are limited and often not accessible to the public. London et al. [18], for example, relied on unpublished data to model impacts from PA-12 powder production in order to build the life cycle inventory for PBF. In this study, Ecoinvent data for nylon 6 were used, given its traceability and data quality documentation. A sensitivity analysis was performed by using the LCIA data from London et al. to see how the impact results are affected by the data source.

- SA3: Transportation of imported products by plane instead of ship.

The main inputs for the PBF process are imported from Europe and the United States by sea. A sensitivity analysis considered the scenario of importing the same inputs by long-distance air transportation, mainly to verify if there is a significant difference in impacts between these scenarios.

- SA4: Number of printed parts needed to fulfill the declared unit.

Considering subsea conditions for the use phase, a sensitivity analysis assumed that the aluminum housing would have a longer lifespan than its PA-12 counterpart. Scenarios considered the need to produce four, five, and six PA-12 housings to meet the requirements of the declared unit for a lifespan of 5 years.

3. Results

The assessment of GWP and CED for the production of one camera housing is presented and discussed, considering additive manufacturing (PBF) and conventional manufacturing (CNC Machining) as production scenarios.

3.1. Global Warming Potential and Energy Demand

Table 5 presents the results (cradle-to-gate) for the chosen impact categories, while Figure 3 displays the relative impacts grouped by process. The analysis of the outcomes revealed that the PA-12 part produced using PBF technology presented superior environmental and energy performance when compared with the metal component manufactured via CNC machining. There was a reduction of around 90% in carbon emissions and 84% in cumulative energy demand, with statistically significant variances between the two scenarios.

Table 5. Environmental and energy performance of one camera housing in the production systems considered.

Manufacturing Process	Reference Flow	GWP	CED
		kgCO ₂ e	MJ
PBF	1.84 kg—PA-12	47.47	906.94
CNC machining	6.40 kg—aluminum alloy	456.82	6249.91

Global warming potential (GWP); cumulative energy demand (CED).

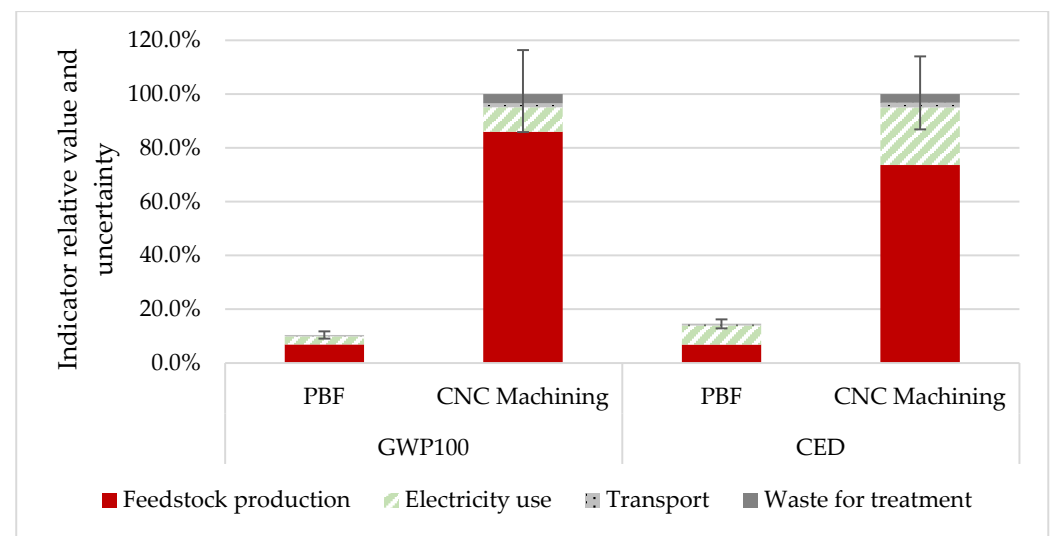


Figure 3. Relative environmental and energy performance contribution per group and overall propagated uncertainty for the production of one camera housing using PA-12 PBF and aluminum CNC machining.

In the case of CNC machining, the impact of aluminum production dominated both considered indicators, accounting for 86% of GWP and 74% of CED. These findings align with the existing literature, which also highlights the significant influence of material production in aluminum machining processes [48]. For instance, Ingarao et al. [49] conducted an evaluation of environmental and energy performance in machining and forming processes, revealing that aluminum production contributed to approximately 96% of carbon emissions and 98% of energy consumption in CNC machining. Similarly, Priarone et al. [50] found similar values for energy consumption, with aluminum production accounting for 98% of the total for the process. The use of electricity emerged as the second-largest contributor to the assessed impacts, accounting for 9% of GWP and 21% of CED. Discrepancies between the outcomes of this research and the literature could be attributed to the quality

of electricity consumption data for the machining stage, which was estimated based on internal reports and personal communication.

In the case of PBF, the production of PA-12 powder was the dominant contribution to the GWP indicator, accounting for 66%, and the second major contribution for CED, accounting for 47%. Electricity use was the second major contribution for GWP, accounting for 29%, and represented the highest impact in terms of CED, accounting for 49%. In general, studies assessing the life cycle of additive manufacturing processes have observed that the major contribution to impacts occurs during the processing stage, mainly due to high electricity demand, irrespective of whether the materials used are metals [15,22] or polymeric materials [20]. London et al. [18], in an LCA study focusing on the PBF process, found that material production and electricity use contributed to 39% and 60% of the total GWP impacts, respectively. The relative lower contribution of electricity use in this study can be explained by Brazil's energy matrix, which has a lower environmental impact compared to the global average since it is mainly based on hydroelectric systems (~62% [51]) [52].

3.2. Sensitivity Analysis

Figure 4 displays the results of the sensitivity analysis conducted for the PBF process parameters. The outcomes from the analysis clearly indicate substantial variations between the different scenarios when compared to the baseline scenario, emphasizing the importance of the chosen process parameters in influencing the environmental and energy performance of the PBF manufacturing process.

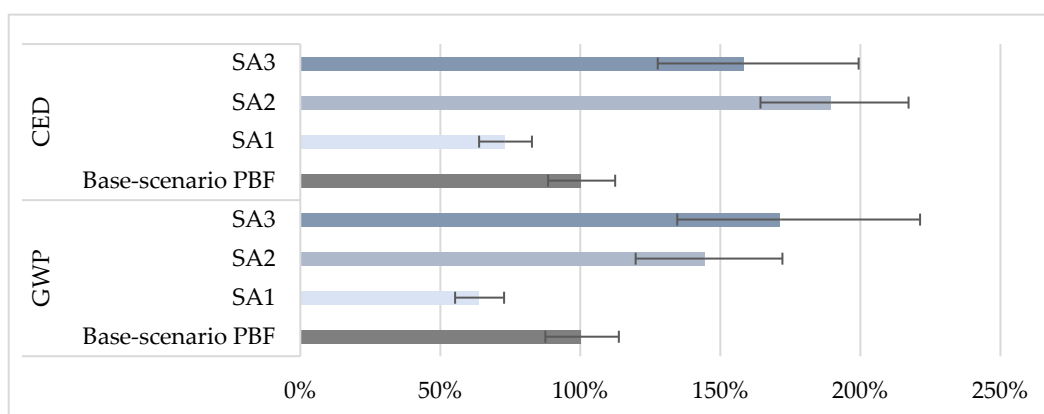


Figure 4. Sensitivity analysis for PBF parameters for the production of one camera housing.

Increasing the proportion of recycled material in the PBF printing chamber by 10% (SA1) resulted in a significant reduction in the environmental impacts of the process. Specifically, there was a 36% decrease in GWP and a 27% decrease in CED. This reduction can be primarily attributed to the diminished impacts resulting from the production and transportation of virgin PA-12 powder.

A study by Riedelbauch et al. [53], where they investigated the properties of PBF-produced parts using PA-12 powder recycled multiple times while varying the proportion of the mixture between virgin and recycled material, concluded that the utilization of 100% recycled material would not compromise the mechanical properties of printed parts for at least five processing cycles. This suggests that adopting a higher percentage of recycled material in PBF manufacturing has the potential to significantly further reduce the process's environmental impact.

The utilization of life cycle impact data for PA-12 powder from London et al. [18] (SA2) led to a considerable increase of 44% in GWP and 89% in CED. Using PA-6 LCIA data as a replacement for PA-12 is often considered an option in LCA studies given the lack of peer-reviewed information [30,31]. However, these results highlight the critical importance of using LCIA data that accurately reflect the conditions of the system under

study, especially considering the high variability in values found in the literature. For example, Kilchert [29], in an LCA report on PA-12 powder production and distribution for additive manufacturing, found a GWP value of 5.70 kg CO₂e per kg, which was 60% lower than the value used by London et al. (15.39 kg CO₂e).

Moreover, a significant concern arises from the lack of quality analysis of sources and the absence of a method for estimating data uncertainties for most of the information available on PA-12 production. Addressing uncertainties is an inherent aspect of LCA studies, and evaluating data quality is crucial, especially in comparative studies [14,54].

The decision to switch from maritime to air transportation (SA3) for importing inputs like PA-12 powder and additives used during printing operations had a substantial impact compared to the baseline scenario, with an increase of 71% in GWP and 58% in CED. The impacts of the transportation phase became equivalent to those of material production, emphasizing the critical role of transportation choices in the overall environmental performance of a manufacturing process.

Figure 5 shows the impacts related to varying the number of PA-12 housings needed to meet the declared unit lifespan of 5 years (SA4), in comparison with the aluminum part.

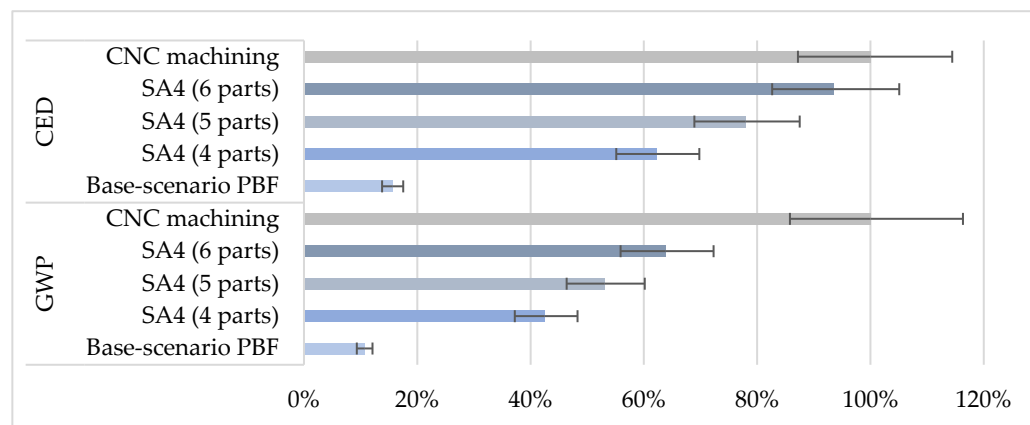


Figure 5. Sensitivity analysis of the reference flow for the declared unit when considering different values for PBF lifespan for the production of one camera housing.

The analysis indicated that the PA-12 housing continued to demonstrate advantages in terms of GWP and CED, even in a scenario where five printing jobs would be required over a 5-year period. This suggests that, despite the need for multiple printing jobs and potentially shorter lifespans, the use of additive-manufactured polymers for metal replacement could be a viable option for sustainable manufacturing practices and reduced environmental impacts.

4. Discussion

The use of additive manufacturing methods for producing aluminum parts may not always provide significant environmental advantages compared to conventional machining processes, even when considering the significant weight reduction [48,55]. For that reason, this study explored an alternative approach, especially for non-structural parts: to replace aluminum with engineered polymer parts produced by additive manufacturing. In this context, the PBF process using PA-12 as the raw material emerges as a noteworthy option, offering several advantages over traditional aluminum manufacturing.

One significant advantage was the weight reduction of the produced part. The PBF plastic part was ~3.5 times lighter than its metal counterpart, which has the potential to reduce impacts caused by fuel and electricity consumption during the use phase. This would be particularly important in the aerospace sector, for example, where fuel efficiency is critical. Of course, this change in material would need to be based on a preliminary analysis of viability, and the product may need to be redesigned to meet the mechanical requirements during utilization.

A lower carbon footprint for the PA-12 part was also observed, even when considering a reduction in lifetime compared with aluminum. Most emissions from the PBF process were related to electricity and PA-12 powder production, which have great potential for reduction in the future. Plastics produced from renewable resources, such as PA-11, could be used to reduce the production chain impacts of plastics while using the same MJF printer. If we consider the lifecycle emissions factors for plastic production from Brehmer [56], changing PA-12 to PA-11 could reduce carbon emissions by up to 40%, although differences in mechanical strength should be considered [57]. Another option could focus on the electricity source, by using renewables such as wind and photovoltaic power. The Brazilian energy grid already benefits itself in terms of carbon intensity when compared to countries dependent on coal or natural gas. However, if we consider the ecoinvent[®] emission factors for Brazil, we could have reductions of 93% and 56% by changing the energy source to 100% wind and 100% photovoltaic, respectively.

Another advantage that can be noted is MJF's lower electricity consumption when compared to conventional aluminum and even metal PBF methods such as SLM [18,58]. As can be seen in Table 6, the MJF process has an 80 % lower specific energy consumption when compared to SLM. Total energy use would also be reduced considering the final weight of the finished component.

Table 6. Specific energy consumption values for additive manufacturing processes considering aluminum and PA-12 as input materials.

Process	Machine Tool	Material	Specific Energy Consumption/ MJ/kg	Source
MJF	HP MJF 5210	PA-12	118.2	This study (2)
MJF	HP MJF 4210	PA-12	98.7	
SLM	Reinshaw AM250	Al-Si	567.2	(3)

(1) This study; (2) London et al. [18]; (3) Faludi et al. [59].

Moreover, polymeric additive manufacturing processes like PBF often result in shorter design and production times compared to traditional metal manufacturing techniques. This faster production turnaround can lead to reduced lead times and potentially lower energy consumption during the manufacturing phase, further contributing to the overall environmental benefits of using polymeric additive manufacturing for non-structural parts.

As the field of additive manufacturing continues to advance and technologies like PBF become more refined, conducting comprehensive environmental assessments becomes crucial. This helps ensure that the potential environmental benefits of such innovative manufacturing processes are accurately understood and that they are appropriately incorporated into sustainable design and production practices.

5. Limitations

It is important to emphasize that these results do not represent a declaration of environmental superiority of one process over the other. Limitations of this study include the use of estimated data for electricity consumption for both PBF and CNC machining processes, which may not fully capture real energy-use variations, and the consideration of only two environmental indicators, potentially overlooking other significant impacts.

Moreover, there is still much uncertainty regarding the lifespan of both materials (aluminum and PA-12) considering the intended use to define a proper functional unit. Further investigation is needed to explore how these materials would react in subsea conditions and which options are available to improve their properties in this case.

Finally, it is essential to note that this analysis solely focused on the cradle-to-gate boundary, meaning it did not consider the impacts during the use and end-of-life phases, which can significantly influence the overall life cycle environmental performance of a product.

6. Conclusions

The results of this study demonstrate that:

- The plastic component manufactured using PBF had better environmental (global warming) and energy performance compared to the metal component manufactured using CNC machining.
- There was a significant reduction of approximately 90% in equivalent carbon emissions and 84% in cumulative energy demand, with statistically significant differences between the two scenarios.
- The impacts of the PBF process exhibited high sensitivity, mainly due to changes in the proportion of virgin/recycled material in the print chamber, the use of different data sources for the life cycle inventory (LCI) of PA-12 production, and the transportation system considered for imported inputs.
- According to the sensitivity analysis, considering the cradle-to-gate boundary, the plastic component produced using PBF would still offer environmental advantages if its lifespan was one-fifth that of the metal component.

These findings provide valuable insights into the environmental and energy performance of the studied scenarios, emphasizing the importance of considering different factors, such as material choice, production processes, and transportation methods, in assessing the sustainability of manufacturing processes.

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Appendix A

Table A1. Inventory of the production of 1 kg of fusion agent, cradle-to-gate lifecycle.

Item	Value	Unit	SD ² _g	Source
Input				
Electricity, medium voltage	0.67	kWh	1.809	(1)
Heat, natural gas	3.34	MJ	1.809	(1)
Carbon black	0.07	kg	1.541	(2); (3)
2-pyrrolidone	0.19	kg	1.541	(2); (3)
Water, deionized	0.74	kg	1.541	(2); (3)

Table A1. Cont.

Item	Value	Unit	SD ² _g	Source
Output				
Reference product				
HP fusion agent	1.00	kg	-	

(1) ecoinvent 3.8 [26] based on dataset: “printing ink production, offset, product in 47.5% solution state | printing ink, offset, without solvent, in 47.5% solution state | APOS, U”; (2) HP [60]; (3)/London et al. [18].

Table A2. Inventory of the production of 1 kg of detailing agent, cradle-to-gate lifecycle.

Item	Value	Unit	SD ² _g	Source
Input				
Electricity, medium voltage	0.67	kWh	1.809	(1)
Heat, natural gas	3.34	MJ	1.580	(1)
Triethylene glycol	0.14	kg	1.541	(2); (3)
2-pyrrolidone	0.04	kg	1.541	(2); (3)
Water, deionized	0.82	kg	1.541	(2); (3)
Output				
Reference product				
HP fusion agent	1.00	kg	-	

(1) ecoinvent 3.8 [26] based on dataset: “printing ink production, offset, product in 47.5% solution state | printing ink, offset, without solvent, in 47.5% solution state | APOS, U”; (2) HP [61]; (3) London et al. [18].

Table A3. Power consumption and time duration of PBF printing processes.

Data	Duration/ h	Power Consumption/ kW	Electricity Use/ kWh	Source
Powder mixing and loading	-	-	1.56	
Processing station setup	0.16	0.3	0.05	(1); (2)
Mixing and loading	0.58	2.6	1.52	(1); (2)
HP Multi-Jet Fusion printing	-	-	108.60	
Printer setup	1.50	0.3	0.45	(2); (3)
Printing process	9.00	12.0	108.00	(2); (3)
Safety cooling	0.50	0.3	0.15	(2); (3)
Cooling and unpacking	-	-	0.57	
Processing station setup	0.16	0.3	0.05	(2); (3)
Unpacking	0.20	2.6	0.52	(2); (3)
Post-processing/cleaning	-	-	1.15	
Cleaning	0.25	3.7	1.15	(4); (5)
Total			111.89	

(1) HP [62]; (2) London et al. [18]; (3) HP [32]; (4) U.S. Department of Energy [63]; (5) Internal communication.

Table A4. Transportation inventory for PBF material inputs.

Material	Transport Mode	Distance/(km)	Origin—Destination	Source
PA-12 powder	Truck	50	Marl, DE to Duisburg Port, DE	(1)
	Container ship	12,642	Duisburg Port, DE to Santos Port, BR	(2)
	Truck	1130	Santos Port, BR to SKA distribution center, BR	(1)
	Truck	3101	SKA distribution center, BR to Salvador, BR	(1)

Table A4. *Cont.*

Material	Transport Mode	Distance/(km)	Origin—Destination	Source
Fusion and detailing agents	Truck	79	Corvalis, EUA to Newport Port, US	(1)
	Container ship	16,146	Newport Port, US to Santos Port, BR	(2)
	Truck	1130	Santos Port, BR to SKA distribution center, BR	(1)
	Truck	3101	SKA distribution center, BR to Salvador, BR	(1)

(1) Google Maps [35]; (2) Searate [36].

Table A5. Power consumption and time duration of CNC machining processes.

Step	Time (1)/ h	Power Consumption (2)/ kW	Electricity Use/ kWh
Setup	16	1.0	16.0
Machining	38	4.4	167.2
TOTAL			183.2

(1) SENAI CIMATEC [34]; (2) Montes [37].

Table A6. Transportation inventory for CNC machining material inputs.

Material	Transport Mode	Distance/(km)	Origin—Destination	Source
Aluminum alloy (AA 6061)	Truck	1 977	GDD Metals, SP, BR to Salvador, BR	(1)
Lubricant oil	Truck	2 095	São Paulo, BR to Salvador, BR	(1)

(1) Google Maps [35].

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